

RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track

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Abstract. The NLPCC 2026 Chinese BabyLM shared task studies data-efficient Chinese language modeling with a fixed pretraining corpus. Its Cognitive Track evaluates word- and sentence-level representations through separate fMRI encoding tasks. We examine whether the two tasks benefit from representations tailored to their linguistic scale. RepDuet combines intermediate contextual states with corpus statistics for word evaluation. It uses upper-layer contextual states with lexical and confidence features for sentence evaluation. All components are learned from the official corpus. RepDuet ranked first in the Cog Track after organizer reproduction and final-layer verification, scoring 0.33965 Cog on the hidden evaluation. In a matched development analysis, the task-specific contextual representation scores 0.327620, compared with 0.325875 for the strongest of 18 shared representations. The margin is 0.001745, with a participant-cluster 95% confidence interval (CI) of [0.000246, 0.003453], and remains positive across all 36 leave-one analyses. The evidence supports task-specific representation extraction while leaving the exact layer choice underdetermined. The model and inference code are publicly available.¹

Keywords: Chinese BabyLM · data-efficient language modeling · fMRI encoding · representation learning

1 Introduction

The NLPCC 2026 Chinese BabyLM shared task evaluates language models trained on 100 million Chinese tokens through language understanding, cognitive alignment, and Hanzi knowledge [14]. The Cognitive Track measures how well learned representations predict word- and sentence-level fMRI responses. Ridge models map token representations to each neural dataset [19]. The two evaluations differ in linguistic scale and neural target, which raises a basic representation question: should both rely on the same encoder state?

¹ <https://huggingface.co/LikC1606/chinese-babylm-cog-a197-strict-best>. RepDuet is the paper name for submission A197.

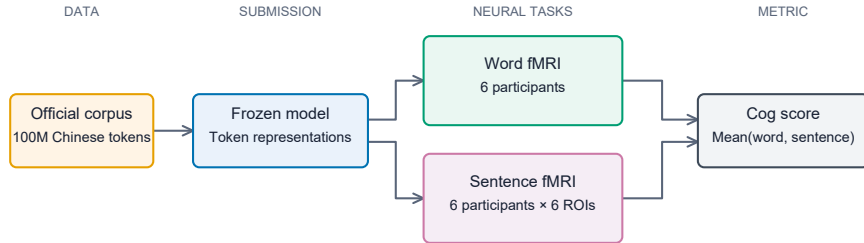


Fig. 1. Evaluation structure of the NLPCC 2026 Chinese BabyLM Cognitive Track. A frozen submission provides token representations to separate ridge-encoding analyses for word and sentence fMRI. Cog is the unweighted mean of the two task scores.

Figure 1 summarizes the Cognitive Track. Models are trained on the official corpus and evaluated by separate linear probes for the word and sentence fMRI datasets. Their scores contribute equally to Cog.

Layerwise studies show that linguistic information changes across encoder depth. Corpus statistics and model confidence provide complementary lexical signals. These observations motivate task-specific representations learned within the same data budget. Development-based selection can exaggerate fine distinctions among layers and feature weights. We therefore examine predictive performance together with the stability of the selected representation.

RepDuet addresses the two fMRI tasks with separate hybrid representations. As Figure 2 shows, word evaluation uses intermediate contextual states and corpus statistics. Sentence evaluation uses upper-layer states together with lexical and confidence features. All components are trained from the official corpus. The organizers reproduced the submitted model and confirmed its first-place Cog Track result.

RepDuet provides an organizer-reproduced model that ranks first in the Cog Track, together with evidence about its task-specific representations. A controlled comparison against 18 shared representations isolates their effect, and participant resampling tests the stability of the observed margin. Development sweeps further reveal broad performance plateaus. The evidence therefore supports a difference in representational depth more strongly than any single layer, placing representation choice and its uncertainty at the center of cognitive benchmark analysis.

2 Related Work

BabyLM introduced controlled small-data pretraining as a test of sample efficiency. The original challenge used 100M- and 10M-word regimes and assessed grammar, transfer, and generalization [16]. Later editions expanded the setting to multimodal and interactive learning [7, 1]. Their findings show that learning

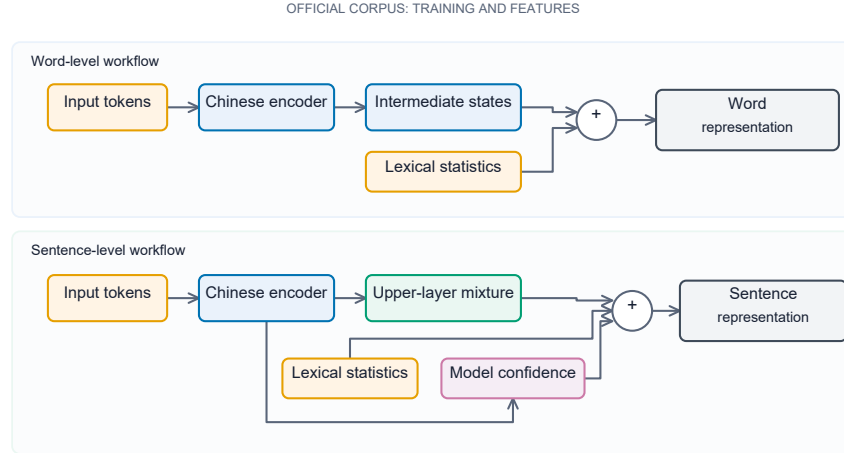


Fig. 2. RepDuet derives task-specific representations for the two fMRI evaluations. Word prediction emphasizes intermediate contextual states; sentence prediction uses upper-layer states with lexical and confidence features. Every feature is estimated from the official corpus.

remains sensitive to model and data choices under limited supervision. Chinese BabyLM brings this program to Chinese and adds Hanzi knowledge and cognitive alignment to standard language evaluation [14].

Neural encoding studies use linear models to relate language representations to brain activity [13, 9]. Their results, together with layerwise analyses of Transformer models [15], suggest that representational depth matters for neural prediction. MulCogBench provides Chinese word- and discourse-level fMRI data for this purpose [19]. RepDuet combines contextual states with random-indexing statistics [10] and model-derived confidence [11]. Our analysis focuses on the division between word and sentence representations and on the uncertainty introduced by development-based model selection.

3 Method

RepDuet maps a token sequence $x = (x_1, \dots, x_T)$ to token-aligned vectors $z = (z_1, \dots, z_T)$. The benchmark task identity selects one of two fixed mappings: a word representation or a sentence representation. Both mappings combine a contextual state with corpus-derived lexical information; the sentence mapping also records model confidence. The resulting vectors are passed directly to the official linear fMRI probes. Figure 2 shows this input–representation–output path.

3.1 Data and Encoders

Every trainable component and lexical feature uses the official Chinese BabyLM corpus [4]. Text is normalized with NFKC and tokenized with a vocabulary that preserves the required CJK characters. Both encoders are initialized randomly. External models, linguistic resources, and CogBench examples are excluded from training and feature estimation. Development scores are used for model selection.

RepDuet contains two compact BERT encoders [6].² For encoder $b \in \{w, s\}$, let $h_{i,\ell}^b$ denote the contextual state of token x_i at layer ℓ . Let r_i^b denote its static lexical feature and u_i its confidence feature. All three quantities use the same tokenizer position, so no pooling or realignment is needed at inference. The task label fixes which mapping is evaluated; the choice does not depend on the text, neural targets, or probe scores. The released model fixes the weights and inference procedure [12]. The organizers reproduced the model from the submitted training code and confirmed its final result [2].

3.2 Task-Specific Representations

The word mapping retains three intermediate states:

$$z_i^w = [h_{i,2}^w; h_{i,3}^w; h_{i,4}^w; r_i^w], \quad (1)$$

where brackets denote concatenation. These states provide contextual evidence from several representational depths. The static feature r_i^w supplies corpus statistics that remain constant across contexts.

The lexical features contain a random-indexing (RI) vector and standardized $\log(1 + c_i)$, where c_i is the token’s context-pair count. RI projects co-occurrence counts into a fixed random basis and normalizes the result by token count [10]. It provides a compact estimate of lexical neighborhood structure.³

The sentence encoder is obtained through checkpoint arithmetic, following the general approach of model soups and task arithmetic [18, 8]. It combines an MLM anchor with continuations trained for next-sentence prediction and static alignment. This construction retains the common MLM solution while introducing directions learned from sentence order and corpus-level lexical geometry. The coefficients are selected on development data and remain fixed during evaluation.⁴

The sentence mapping uses an upper-layer mixture $\bar{h}_i^s$ and forms

$$z_i^s = [\bar{h}_i^s; r_i^s; u_i]. \quad (2)$$

Here r_i^s has the same construction as the word lexical feature, with a wider context window. The vector u_i is computed by an internal MLM and summarizes

² The tokenizer has 16,000 entries; each encoder has 8 layers and hidden size 384.

³ Statistics are estimated from 200,000 corpus sequences. The word and sentence RI vectors have 256 and 384 dimensions, respectively.

⁴ For anchor S_a , next-sentence continuation S_n , and static-alignment continuation S_r , we use $S_b = S_a - 0.075(S_n - S_a)$ and $S = S_b - 5.1(S_r - S_a)$.

the observed token’s likelihood, predictive entropy, leading probability, and prediction margin. These values expose how strongly the model supports each token in context. Development data determine the layer mixture and feature scales, so our conclusions concern the submitted model and its evaluation.

3.3 Training

Training begins with masked language modeling and then separates into word and sentence branches. Replaced-token detection sharpens token-level discrimination in the word encoder [5]. Next-sentence prediction supplies sentence-order supervision to the sentence encoder.

The static-alignment continuation links contextual sentence states to lexical geometry learned from the official corpus. For each sequence, final contextual states and static token embeddings are separately mean-pooled over non-special tokens. A cosine loss aligns the two pooled vectors. A mean-squared-error loss between the normalized student vector and a frozen internal checkpoint limits drift from the MLM representation. The continuation jointly optimizes these terms with the MLM loss. Auxiliary prediction heads are removed before fMRI evaluation, and the frozen encoders emit the vectors in Eqs. 1–2. Each selected model was trained once. Complete schedules and random seeds accompany the source.⁵

4 Evaluation Protocol

Official standing follows the organizer workbook, and the released model reports the hidden-evaluation score [2]. Controlled comparisons use a fixed June 11 development rerun. Ridge selection and score reporting use the same local partition, so all local results are development estimates [3].

The word evaluator fits ridge probes for six subjects and averages the top 10% trial-wise voxel correlations. Sentence fMRI averages standardized prediction–response correlations over 36 region-of-interest–subject cells. Cog is the unweighted mean of the two fMRI scores, following the task specification [14, 19].

The original ridge grid reached its upper boundary for every word participant and 30 of 36 sentence cells. For the feature analysis in Table 3, we recomputed the probes in float64 and extended the ridge grid.⁶ The shared-representation comparison retains the original CogBench protocol.

To isolate task-specific contextual extraction, we remove all appended features and fix the representation width at 384. The comparison includes layers 1–8 from each encoder, the mean of word-encoder layers 2–4, and the selected sentence-layer mixture. Each control supplies one representation to both fMRI tasks. RepDuet supplies word-encoder layer 4 to word evaluation and the sentence mixture to sentence evaluation. Tokenization, model weights, evaluation data, and ridge fitting remain fixed across all 19 conditions.

⁵ See `analysis/training_configuration.csv` in the source archive.

⁶ The expanded grid reaches 10^8 .

Uncertainty is estimated with 100,000 task-stratified participant-cluster bootstrap samples. Model differences are paired within each task, and sentence regions remain grouped by participant. We compute intervals against the observed best shared representation and against the best shared representation reselected in each sample. A complementary analysis removes one participant from each task in all 36 combinations. These procedures measure participant stability on the development data. They leave training variation and model selection uncertainty unresolved.

5 Results

5.1 Competition Result and Development Comparison

Table 1 reports the competition result and selected development stages. The organizer excluded the two higher raw entries from the Cog award: XJTTLU-SeedLM evaluated only open tasks and data, and Alice did not use its final layer. Deepscientist was ranked first after reproduction and final-layer verification [2].

Table 1. Competition result and development comparison. Panel (a) reproduces the organizer workbook; † marks entries excluded from the Cog award. Panel (b) reports selected development stages under one local protocol. Official values retain the organizer’s precision; local values use four decimals because the stage differences are below 0.001. Dashes denote unavailable subtask scores [2, 12].

System	word fMRI	sentence fMRI	Cog	Record / change
<i>(a) Official leaderboard and released model</i>				
XJTTLU-SeedLM [†]	–	–	0.3410	Raw #1; open-only evaluation
Alice [†]	–	–	0.3397	Raw #2; non-final layer
RepDuet	–	–	0.3396	Raw #3; reproduced #1
SMI	–	–	0.3378	Raw #4
CogTeam	–	–	0.3376	Raw #5
RepDuet card	0.5655	0.1138	0.33965	Hidden Evaluation
<i>(b) Selected stages on open development</i>				
Interpolation	0.5626	0.1021	0.3323	Combined sentence checkpoints
Confidence v1	0.5628	0.1032	0.3330	Initial confidence features
Confidence v2	0.5628	0.1033	0.3330	Later confidence model
RepDuet	0.5628	0.1034	0.3331	Selected representation

RepDuet scored 0.3396 in the organizer workbook and ranked first among eligible Cog systems after verification. The released model reports 0.33965 on the hidden evaluation. Development performance rose by 0.000725 after checkpoint interpolation and by 0.000038 in the final step, indicating saturation near the selected model. Table 2 provides the controlled comparison used to study task-specific representations.

5.2 Task-Specific versus Shared Representations

Table 2 compares task-specific contextual extraction with 18 shared representations. Appended lexical and confidence features are removed, and every condition has 384 dimensions.

Table 2. Matched development comparison of task-specific and shared contextual representations. Scores are rounded to three decimals. Cog is the mean of word and sentence fMRI scores. $\star$ marks the strongest shared representation.

Representation	word fMRI	sentence fMRI	Cog
Task-specific	0.560	0.095	0.328
<i>Shared word-encoder states</i>			
L1	0.558	0.078	0.318
L2	0.560	0.082	0.321
L3	0.560	0.083	0.322
L4	0.560	0.083	0.322
L5	0.558	0.085	0.322
L6	0.558	0.088	0.323
L7 $\star$	0.558	0.093	0.326
L8	0.558	0.090	0.324
mean(L2–L4)	0.560	0.083	0.322
<i>Shared final sentence-encoder states</i>			
L1	0.558	0.080	0.319
L2	0.559	0.084	0.321
L3	0.557	0.085	0.321
L4	0.558	0.085	0.321
L5	0.556	0.088	0.322
L6	0.557	0.090	0.324
L7	0.554	0.095	0.324
L8	0.555	0.093	0.324
0.98L7 + 0.02L8	0.554	0.095	0.324

The task-specific representation combines word-encoder layer 4 with the sentence-layer mixture. The strongest shared representation is word-encoder layer 7 at 0.325875, giving a Cog margin of 0.001745. Word fMRI varies little across intermediate layers, while sentence fMRI improves toward the upper layers. States from the sentence encoder show the converse pattern. The two fMRI tasks therefore favor different regions of the representation space.

The precise word layer is weakly identified. Layers 2–5 differ by only 0.000358 Cog, and layer 3 slightly exceeds layer 4 in both fMRI scores. Layer 4 is retained from the submitted model. The evidence identifies task separation more clearly than a unique word layer.

Against word-encoder layer 7, the participant-cluster bootstrap gives 95% CI [0.000246, 0.003453]; 99.0% of draws have $\Delta > 0$. Reselecting the best shared state in every draw gives [0.000246, 0.003335], again with 99.0% positive draws. The second interval accounts for variation in the strongest shared representation within the defined set. Both intervals remain conditional on development-based selection.

In the exact leave-one grid, the margin stays positive in all 36 combinations (range 0.000731–0.002493; median 0.001797). Word-encoder layer 7 is the

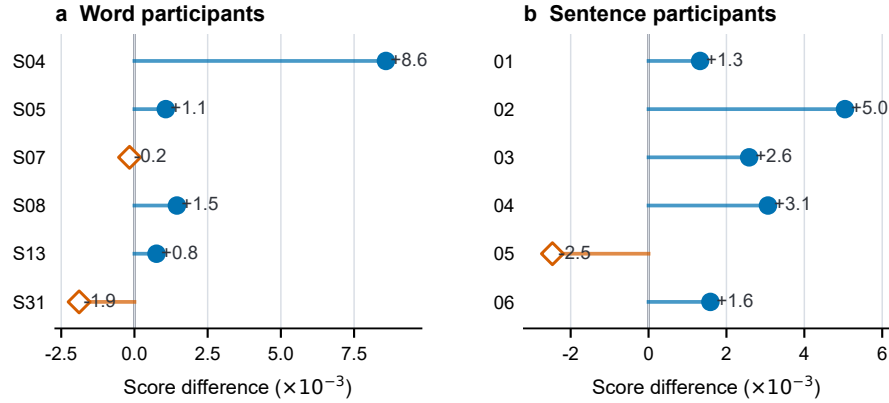


Fig. 3. Participant-level positive and negative cases relative to the strongest shared representation, word-encoder layer 7. Points show the RepDuet score minus the shared score; circles denote positive differences and open diamonds denote negative differences. Sentence values average the six regions for each participant. All values come from the matched development analysis.

best shared competitor in 35 combinations and sentence-encoder layer 7 in one. Thus, no single recorded participant determines the sign; the check nevertheless remains post-selection and development-only.

The aggregate advantage is not uniform across participants. As Figure 3 shows, the task-specific representation is higher for four of six word participants and five of six sentence participants. The negative cases in both tasks motivate participant-level uncertainty analysis and caution against interpreting the mean as a universal gain.

6 Analysis

6.1 Selection Stability

The development sweeps form broad performance plateaus. Confidence variants span less than 0.00014 sentence fMRI, and the checkpoint sweep spans 0.000341. The submitted setting lies within 2×10^{-6} of the best observed score. Development data thus identify a high-performing region while leaving its precise location weakly determined. Figure 4 summarizes these sweeps.

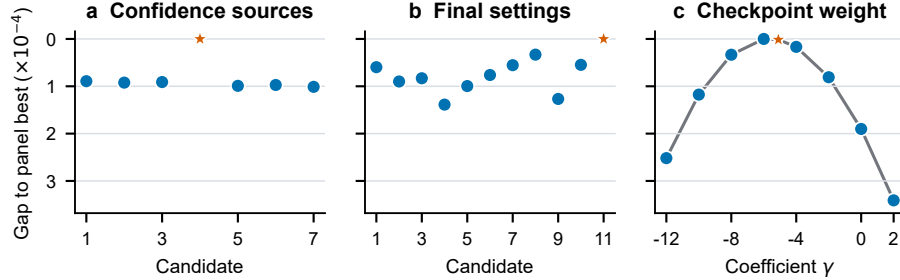


Fig. 4. Development sensitivity of confidence features and checkpoint arithmetic. Each panel reports the distance from the best sentence-fMRI score; orange stars mark the selected settings. The small score differences indicate broad performance plateaus.

6.2 Auxiliary Feature Sensitivity

We estimate local feature sensitivity by setting each auxiliary block to zero and refitting the ridge probes on an expanded float64 grid. This intervention measures the response of the trained representation to feature removal.

Table 3. Sensitivity to auxiliary feature removal. Scores use three decimals; Δ Cog uses four to retain small effects and is measured against the float64 baseline after ridge refitting.

Variant	word fMRI	sentence fMRI	Cog mean	Δ Cog
Baseline rerun	0.563	0.104	0.333	0.0000
Static removed	0.561	0.101	0.331	-0.0022
Confidence removed	0.563	0.103	0.333	-0.0001
Context only	0.561	0.101	0.331	-0.0021

The float64 rerun removes the numerical warnings observed in the original probe. Removing the static block lowers Cog by 0.002225; removing confidence features lowers it by 0.000106. The blocks differ in dimension and scale, and their effects interact through ridge refitting. These values quantify local sensitivity of the selected representation. They do not estimate the effect of retraining without each feature family.

7 Research Process and Limitations

7.1 Use of DeepScientist

DeepScientist [17] assisted the iterative study by tracking hypotheses, launching approved experiments, and organizing results. The authors defined the data constraints, chose the analyses, selected the submitted model, verified the scores,

and wrote the scientific interpretation. DeepScientist had no access to fMRI labels or item-level evaluation feedback.

7.2 Reproducibility and Limitations

The organizers received the training code, tokenizer, computational environment, and model history, and independently reproduced the submission [2]. The public release contains the model and inference code. The accompanying material provides training schedules, participant-level scores, all 18 shared-representation results, uncertainty estimates, and file hashes.

Model selection considered at least 471 development-scored candidates without an independent test partition or repeated training seeds. Participant resampling occurs after selection and does not capture stimulus variation, training variation, or performance on new data. Auxiliary-feature removal is performed after training. The findings therefore characterize the selected model on the available benchmark and do not establish human-like cognition.

8 Conclusion

RepDuet used task-specific hybrid representations for word- and sentence-level fMRI encoding. It ranked first in the NLPCC 2026 Chinese BabyLM Cog Track after organizer reproduction and achieved a hidden-evaluation score of 0.340. In a matched comparison with 18 shared representations, it improved Cog by 0.002. Participant bootstrap and leave-one analyses retained the direction of this difference.

The results showed that word and sentence prediction favored different representational depths, while nearby configurations performed similarly. The evidence supported task-specific extraction more strongly than any exact layer choice. The study thus identified representation choice and selection uncertainty as central concerns in cognitive benchmark analysis.

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